Related-Work Table - BUSI-WHU (CSE475 Task 1)

Group Group08 | Members: Maharun Nesa Tanni (2023-1-60-190), Kh. Tania Akter Tamanna (2023-2-60-070) | Track 2 (GCN) / Track 3 (CNN+Attention) | 6 papers, 2024-2025. Items in [brackets] were not confirmed from the sources I could open: check them on the paper page before submitting.

#	Title	Authors	Year / Venue	Dataset	Application	Method / model	Metrics (as reported)	Strengths	Limitations	Research gap	Relation to our work	Track 2: graph construction + GCN type	Track 3: attention type
1	Dual branch segment anything model-transformer fusion network for accurate breast ultrasound image segmentation (DSATNet)	[verify authors on paper page]	2025 / Medical Physics	BUSI-WHU (927 imgs: 560 benign, 367 malignant) + BUSI	Breast US tumour segmentation	Dual-branch Swin Transformer + Segment Anything Model with multi-level feature fusion	mIoU, HD95, effect size, p-value; BUSI: mIoU 82.83%, HD95 71.13 px	Uses BUSI-WHU; statistical testing; code public (github.com/Skylanding/DSATNet)	Segmentation only (no benign/malignant classification); heavy SAM+Swin model	No classification or graph/region reasoning on BUSI-WHU	Same dataset; gives a segmentation SOTA reference and shows BUSI-WHU is usable	None (no graph)	Self-attention (Swin) + foundation-model prompts: hybrid
2	DFLNet: Disentangled feature learning network for breast cancer ultrasound image segmentation	[verify authors on paper page]	2025 / Digital Signal Processing	Dataset-B, BUSI, BUSI-WHU, TN3k	Breast US tumour segmentation	Disentangles low-level texture and deep semantic features; stacked spatial + cross-attention to suppress pseudo-lesion areas	Multiple segmentation metrics; reported better than 12 compared methods	Introduces/benchmarks BUSI-WHU (927 imgs, 774 patients); 4-dataset evaluation	Segmentation only; patient-level split details not confirmed	No classification, no graph, no explainability study	Source of the BUSI-WHU dataset; defines the segmentation baseline task	None (no graph)	Spatial attention + cross-attention (residual)
3	Enhanced multi-scale selective attention U-Net for breast ultrasound image segmentation	Wang et al. [verify full list]	2025 / Medical Physics	BUSI (452/110/195 split) + BUSI_WHU	Breast US lesion segmentation	U-Net with multi-scale selective attention and spatial-kernel convolution (SKConv) fusion	mDice, mIoU; BUSI test mDice 68.33%	Attention U-Net family (close to our Attention U-Net baseline); two datasets	Benign and malignant merged into one class; random split (patients not grouped)	No class-aware analysis; no graph	Direct reference for U-Net vs Attention U-Net comparison	None (no graph)	Multi-scale selective (kernel) attention
4	Node-Based Graph Convolutional Network With SLIC Method for Breast Cancer Ultrasound Images Classification	Trang, Ting, Vuong, Ting	2024 / IEEE Access, vol. 12	BUSI, BUS-BRA	Breast US classification (benign / malignant / normal)	SLIC superpixels as nodes; edges by colour similarity + spatial proximity; multi-layer GCN; node voting for image label	Classification metrics on both datasets (see paper)	Real image-to-graph pipeline; studies number of connections	Superpixels cover whole image (background nodes); BUSI has duplicate/patient issues; not tested on BUSI-WHU	No tumour-mask-guided graph; no leakage-aware evaluation	Closest to Track 2: shows image-to-graph GCN works on breast US	Superpixel (SLIC) graph, colour+spatial edges, GCN	None
5	Malignancy pattern analysis of breast ultrasound images using clinical features and a graph convolutional network	Montaha, Azam, Bhuiyan, Chowa, Mukta, Jonkman	2024 / Digital Health	BUSI-type breast US datasets (A, B, A+B)	Benign vs malignant classification	Graph of clinical markers (shape, orientation, margin, surrounding tissue); Pearson-correlation edges; GCN vs GAT	Accuracy 91.03% / 94.37% / 89.62% on supplementary datasets A / B / A+B	Compared with GAT, 1D-CNN, transfer learning, 10 ML and 3 ensemble models	Handcrafted markers, not image regions; nodes are features not places	Graph is not spatial; no segmentation-to-graph link; no CV significance test reported	Shows graph classification benefit; motivates mask-derived shape features	Feature-node graph, Pearson edges, GCN (also GAT)	None
6	HyFormer-Net: A Synergistic CNN-Transformer with Interpretable Multi-Scale Fusion for Breast Lesion Segmentation and Classification in Ultrasound Images	Rahman, M. A.	2025 / arXiv: 2511.01013	BUSI (+ external set for transfer)	Segmentation + classification, XAI	EfficientNet-B3 + Swin dual-branch encoder, multi-scale fusion, attention gates	BUSI Dice 0.761 +/- 0.072, classification acc 93.2%; attention gates +5.9% Dice; zero-shot transfer Dice 0.058	Ablation of attention; explainability; domain-shift test	Preprint (not peer reviewed); BUSI duplicates; no BUSI-WHU	Cross-dataset generalisation poor	Track 3 reference for ablation + Grad-CAM style analysis	None	Attention gates + Swin self-attention (hybrid)

Research gap (from these papers): (1) BUSI-WHU papers (DFLNet, DSATNet) do segmentation only. None of the papers I found benchmark benign-vs-malignant classification on BUSI-WHU. (2) Image-to-graph GCN work (Trang et al.) uses BUSI/BUS-BRA with whole-image SLIC superpixels, not tumour-mask-guided region graphs. (3) Little leakage-aware evaluation (patient-grouped folds, significance tests) is reported; BUSI has no patient IDs, while BUSI-WHU comes with a patient-info file, so grouped splits are

possible. (4) Explainability (Grad-CAM/LIME/GNNExplainer) is rarely reported alongside attention or graph models. **Our contribution:** a patient-grouped, leakage-free comparison of CNN baselines, attention CNN and a mask-guided region-graph GCN on BUSI-WHU with ablation, 5-fold CV, significance test and XAI.